



UTM
UNIVERSITI TEKNOLOGI MALAYSIA

SECI1143-02 Probability & Statistical Data Analysis
Assignment 2

Lecturer: Dr. Noorfa Haszlinna Binti Mustaffa

Group Members:

Name	Matric Number
Goe Jie Ying	A23CS0224
Lam Yoke Yu	A23CS0233
Teh Ru Qian	A23CS0191
Tan Yi Ya	A23CS0187

1 (a) Mean

Class interval	Midpoint	Frequency	$f_i X_i$
150 - 160	155	12	1860
160 - 170	165	20	3300
170 - 180	175	5	875
180 - 190	185	3	555
total		40	6590

$$\text{mean, } \bar{x} = \frac{6590}{40} = 164.75$$

(b) Median

Class interval	Frequency	Cumulative Frequency
150 - 160	12	12
160 - 170	20	32
170 - 180	5	37
180 - 190	3	40
Total	40	

$$\begin{aligned} N &\div 2 \\ &= 40 \div 2 \\ &= 20 \\ \text{median class} \\ &= 160 - 170 \end{aligned}$$

$$\begin{aligned} \text{median} &= L + \frac{\frac{N}{2} - (cf_p)}{f_{\text{med}}} (w) \\ &= 160 + \frac{\frac{40}{2} - 12}{20} (10) \\ &= 164.0 \end{aligned}$$

(c) mode

$$\begin{aligned} &= l + h \times \left[\frac{(f_1 - f_0) \div (2f_1 - f_0 - f_2)}{1} \right] \\ &= 160 + 10 \times \left[\frac{(20 - 12) \div (2(20) - 12 - 5)}{1} \right] \\ &= 163.48 \end{aligned}$$

(d) Modal class : $160 \leq x < 170$

Question 2

$$(a) \text{ mean} = \frac{75+80+82+85+85+85+88+90+90+92}{10} = 85.2$$

$$\text{median} = \frac{85+85}{2} = 85$$

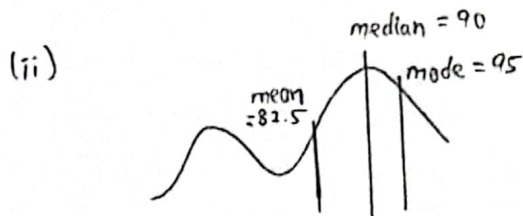
$$\text{mode} = 85$$

(b) According to the values, the performance is quite consistent at about 85. Mean would be more appropriate represent the summary as there are no extreme values and can show the overall performance of the entire class.

$$(c) (i) \text{ mean} = \frac{55+65+65+70+85+95+95+95+100+100}{10} = 82.5$$

$$\text{median} = \frac{85+95}{2} = 90$$

$$\text{mode} = 95$$



(iii) The performance are not quite equal as some performed perfectly while some may struggle to perform well. Compared to the previous one, although the median and mode are greater in value, the mean is reduced.

3. (a) (i) Range = $40 - 25$
 $= 15$

(ii) Variance, $\sigma^2 = \frac{230}{12}$
 $= 19.167$

(iii) standard deviation, $\sigma = \sqrt{\sigma^2}$
 $= \sqrt{19.167}$
 $= 4.378$

x	25	30	28	35	32	27	40	38	33	36	31	29
$(x - \bar{x})$	-7	-2	-4	3	0	-5	8	6	1	4	-1	-3
$(x - \bar{x})^2$	49	4	16	9	0	25	64	36	1	16	1	9

$$\sum (x - \bar{x})^2 = 49 + 4 + 16 + 9 + 0 + 25 + 64 + 36 + 1 + 16 + 1 + 9 = 230$$

(b) The range (15k) is showing the extent of variability of the performance.

The variance (19.167k) and standard deviation (4.38k) showing the fluctuations in sales performance throughout the year.

(c) A higher variance and standard deviation means high variability in sales.

Understanding and acknowledgement of the variance and standard deviation means the company can monitor their performance, for example if value have gone lower means stability in sales. Furthermore, company can do inventory management as high variability may require more stock to prevent stockout during high demand variability.

4. (a) $\mu=50, \sigma=10$

$$\begin{aligned} \mu &= np & \sigma &= npq & p &= 1-q & \text{Percentage} &= 0.8 \times 100\% \\ \therefore np &= 50 & 10 &= (50)q & &= 1-0.2 & &= 80\% \\ & & q &= 0.2 & p &= 0.8 & & \therefore 80\% \end{aligned}$$

$$\begin{aligned} \text{(b)} P(37 < X < 65) &= P\left(\frac{37-50}{10} < Z < \frac{65-50}{10}\right) \\ &= P(-1.3 < Z < 1.5) \\ &= P(Z < 1.5) - P(Z < -1.3) \\ &= 0.93319 - 0.09680 & \therefore 0.836 \\ &= 0.83639 \\ &\approx 0.836 \end{aligned}$$

$$\begin{aligned} \text{(c)} P(X < 20) &= P\left(Z < \frac{20-50}{10}\right) \\ &= P(Z < -3) \\ &= 0.00135 & \therefore \text{RM } 400 \\ 0.00135 \times 1000 &= 1.35 \\ &\approx 2 \text{ people} \\ 2 \times \text{RM } 200 &= \text{RM } 400 \end{aligned}$$

$$\begin{aligned} \text{(d)} P(Z \geq x) &= 0.05 \\ P(Z < x) &= 0.95 & \therefore 66.45 \\ Z &= 1.645 \\ \frac{X-50}{10} &= 1.645 \\ X &= 66.45 \end{aligned}$$

Question 5.

$$b(x, 6, 0.25)$$

(a) X = number of correct answers a student gets on the exam.

(b)

$X = x$	0	1	2	3	4	5	6
$P(X)$	0.1780	0.3560	0.2966	0.1318	0.0330	0.0044	0.0002

$$(c) \mu = 6 \times 0.25 = 1.5$$

$$(d) P(X \geq 3) = 0.1318 + 0.0330 + 0.0044 + 0.0002 \\ = 0.1694 \approx 0.169$$

$$(e) P(X = 4) = (0.75)(1 - 0.75)^3 \\ = 0.012$$

Question 6

$$b^*(x, 4, 0.70)$$

(a) Negative binomial distribution.

$$(b) \sigma^2 = \frac{1-p}{p^2} = \frac{1-0.70}{0.70^2} \\ \sigma^2 = 0.6122 \\ \sigma = 0.782$$

$$(c) b^*(6, 4, 0.70) = \frac{{}^5C_3 (0.30)^4 (0.70)^2}{\rightarrow 0.0397}$$

$$(c) b^*(6, 4, 0.70) = {}^5C_3 (0.70)^4 (0.30)^2 \\ = 0.216$$

$$(d) b(7, 12, 0.70) = {}^{12}C_7 (0.70)^7 (0.30)^5 \\ = 0.158$$